Preliminary Fin Drag Study

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Repository: `https://github.com/sidd-somadder/rogue-fin-design`

The fin redesign aims to reduce planform area, and with it fin drag, while keeping a static stability margin of at least 1.5 calibers from rail exit to motor burnout [4]. Comparing candidate planforms therefore needs a drag estimate that can resolve differences of a few percent between fins of similar size. This document describes the skin-friction estimate used in the planform trade studies, why it is used instead of simulated apogee, and how it was verified. See `drag_estimate.py` in repository.

Simulated apogee in OpenRocket was the first metric considered. However, the planform simulations are run with turbulence, and in Iteration 2 the apogee differences between candidates fell within the run-to-run scatter (about 17 ft), so apogee could not rank the planforms. An independent estimate from boundary-layer theory is used instead. Its results are not exact predictions: they establish the magnitude of fin skin-friction drag, rank candidates relative to one another, and provide a check on the drag OpenRocket reports.

Primarily, the one-seventh power law derivation is used for the coefficient of friction at a point along a flat plate. The velocity profile in the turbulent boundary layer is assumed to follow a one-seventh power curve:

$$\frac{u}{U_\infty} = \left(\frac{y}{\delta}\right)^{\frac{1}{7}}$$

Combining this with von Kármán's momentum integral equation and an empirical wall-shear relation, the local skin-friction coefficient is a function of the distance x along the plate [1]:

$$c_f(x) = \frac{0.0592}{Re_x^{0.2}}$$

where Re_x is the local Reynolds number. Averaging over a plate of length L,

$$C_f = \frac{1}{L}\int_0^L c_f(x)\,dx = \frac{0.0592\,\nu^{0.2}}{U_\infty^{0.2}L}\int_0^L x^{-\frac{1}{5}}\,dx = \frac{0.0592}{Re_L^{0.2}L^{0.8}}\cdot\frac{5}{4}L^{0.8} \approx \frac{0.074}{Re_L^{0.2}},$$

which is generally applicable for $5\times10^5 \le Re_L \le 10^7$.

Form factor. A flat plate has no thickness, so the friction coefficient is multiplied by Raymer's form factor for lifting surfaces [2], which accounts for the higher local velocities over a thick section:

$$FF \approx 1 + \frac{0.6}{(x/c)_m}\left(\frac{t}{c}\right) + 100\left(\frac{t}{c}\right)^4$$

where $(x/c)_m = 0.3$ is the chordwise location of maximum thickness and $t = 0.25$ in is the total fin thickness (core plus skins). Since the thickness is fixed, t/c and therefore FF grow as the chord gets shorter. The location of max thickness was held constant since cross-section differences should not affect comparing planforms and this study is only an estimate rather than providing concrete numbers.

Strip discretization. The chord of a cropped delta varies linearly along the span,

$$c(y) = C_r - (C_r - C_t)\,\frac{y}{s},$$

so a single plate length does not represent the whole fin. The fin is divided into n spanwise strips of width $\Delta y = s/n$, and each strip is given its own Reynolds number $Re_i = Vc_i/\nu$, friction coefficient $C_{f,i}$, and form factor FF_i. The skin-friction drag on one fin is then the sum over the strips, with both faces wetted:

$$\begin{aligned} D_f &\approx C_f \cdot q \cdot S_{wetted} \cdot FF \\ &\approx q_{max}\sum_{i=1}^{n} C_{f,i}\,FF_i\,(2c_i\Delta y) \end{aligned}$$

The sum is converged at $n = 100$ strips. For a single section, the same model gives a two-dimensional drag coefficient based on chord, $c_d = 2C_f\,FF$, which is the quantity compared against XFOIL below.

Flow conditions. Drag is assessed at the worst case, maximum dynamic pressure: $V = 312\,\mathrm{ft/s}$ for the sub-scale, with sea-level density $\rho = 2.377 \times 10^{-3}\,\mathrm{slug/ft^3}$ and kinematic viscosity $\nu \approx 1.6 \times 10^{-4}\,\mathrm{ft^2/s}$. The boundary layer is assumed fully turbulent from the leading edge, so the estimate is an upper bound; any laminar run would lower the drag. The sub-scale reaches $M = 0.28$, so compressibility corrections are neglected.

Assumptions.

(1) Fully turbulent flow overestimates absolute drag, and so the numbers computed by this method is only an estimate is used to assess relative performance under identical assumptions.

(2) At 312 ft/s, chords shorter than about 3 in fall below $Re = 5 \times 10^5$, so the tip strips of the sub-scale fins lie slightly outside the correlation's range, where fully turbulent flow is least likely.

(3) The form factor captures thickness, but not the shape of the leading and trailing edges. Pressure drag from blunt or chamfered edges is not modeled, so cross-sections are compared separately with XFOIL and CFD (see the cross-section study).

(4) Interference drag at the fin–body junction as well as induced drag effects are not included.

(5) The full-scale rocket reaches $M \approx 0.67$, where this assumption no longer holds for absolute values. Raymer's turbulent friction correlation carries a factor $(1+0.144M^2)^{-0.65} \approx 0.96$ at $M = 0.67$, and his full lifting-surface form factor includes a Mach and sweep factor, $1.34\,M^{0.18}(\cos\Lambda_m)^{0.28}$ [2]. At a fixed Mach number these scale all candidates nearly equally, so relative rankings still hold, so for simplicity compressibility effects are neglected. Future studies will require a more precise flutter estimation method.

Verification & Key Observations

The strip sum was checked for convergence with the number of strips; $n = 100$ is converged. As an independent check on the magnitude, the NACA 0005 section was run in XFOIL viscous [3] at $Re = 8.3 \times 10^5$ with transition forced at 1% chord to approximate a fully turbulent boundary layer. XFOIL gave $c_d = 0.00979$, within about 8% of the flat-plate estimate for the same section and Reynolds number.

Drag falls more slowly than area. Across all three iterations, the estimated drag reduction is consistently about 70% of the planform area reduction:

Iteration	Baseline → selected (root × tip × height, in)	Area change	Drag change	Ratio
1	9×3×4 → 7×3×4	−16.7%	−12.0%	0.72
2	9×3×4 → 5.5×3×4.25	−24.8%	−16.4%	0.66
3	10×2×5 → 7.5×2.5×5	−16.7%	−11.9%	0.71

The reason is the chord dependence of each strip. Its drag is proportional to $c_i\,C_{f,i}\,FF_i$, and since $C_f \propto Re^{-0.2} \propto c^{-0.2}$,

$$D_i \propto c_i^{0.8}\,FF\left(\frac{t}{c_i}\right)\Delta y.$$

Shortening the chord lowers the drag only as $c^{0.8}$, and the form factor rises at the same time because the thickness is fixed. Shortening the span instead removes whole strips, so the drag falls in proportion to the span. This is why, in Iteration 3, candidate d (6×2×5.5, 22.0 $\mathrm{in^2}$) has a smaller drag reduction (−17.0%)

than candidate b ($7\times2\times5$, $22.5\,\text{in}^2$, -18.7%) despite having less area: it reaches its area with shorter chords and a longer span. For a given area, cutting chord is less effective at reducing drag than cutting span. Span, however, is what drives stability, so drag and stability pull the planform in opposite directions.

Planform shape at fixed area. In Iteration 1, at a fixed area of $20\,\text{in}^2$ and a fixed cross-section, the estimated drag varied by less than 0.7% across planforms from an $8\,\text{in}/2\,\text{in}$ cropped delta to a rectangle. The estimate therefore distinguishes candidates by area and by how that area is split between chord and span, not by planform shape alone.

References

[1] J. D. Anderson, Jr. and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. New York, NY: McGraw-Hill, 2024.

[2] D. P. Raymer, *Aircraft Design: A Conceptual Approach*, 6th ed. Reston, VA: American Institute of Aeronautics and Astronautics, 2018.

[3] M. Drela, "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils," in *Low Reynolds Number Aerodynamics*, T. J. Mueller, Ed., Lecture Notes in Engineering, vol. 54. Berlin: Springer, 1989, pp. 1–12.

[4] Friends of Amateur Rocketry, *FAR-OUT: Friends of Amateur Rocketry – Oxidizers Uninhibited Tournament, 2026-27 Official Competition Rules and Requirements*, Rev. 2026-08-31, Aug. 2026. [Online]. Available: https://static1.squarespace.com/static/64cc93ec537b6a2fa74f96ba/t/6aa1ff516daf985f5ebb7ae4/1789001554847/FAR-OUT_Rules_Requirements_2026-09-09.pdf (accessed Oct. 10, 2026).